# Shangyong Shi

**Pim Postdoctoral Fellow**

Johns Hopkins University, Department of Earth and Planetary Sciences, Baltimore, Maryland

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## Research Interests

**H**ydroclimatology, **P**recipitation phase partitioning, **S**now hydrology, **R**emote sensing of precipitation, **E**xtreme precipitation, **M**achine learning, **C**limate change

## Education

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|---|---|---|
| 2021 - 2024 | *Ph.D.* | Meteorology, **Florida State University** \| Advisor: Guosheng Liu.<br>Dissertation: On the Sensitivity of Precipitation Phase to Temperature |
| 2018 - 2020 | *M.S.* | Meteorology, **Florida State University** \| Advisor: Guosheng Liu. |
| 2014 - 2018 | *B.S.* | Atmospheric Sciences, **Nanjing University**, China. |

## Professional Experiences

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| 2026.5 - 2026.8 | *Visiting Scholar* | **NSF NCAR** Research Applications Lab \| Host: Cenlin He |
| 2024.9 - now | *Pim Postdoc Fellow* | **Johns Hopkins University** \| PI: Benjamin Zaitchik |
| 2023.6 - 2024.5 | *Research Intern* | **University of Maryland**, Cooperative Institute for Satellite Earth System Studies \| PI: Huan Meng, Yongzhen Fan<br>Improving the machine learning algorithm for orographic snowfall retrieval from satellite passive microwave sensors. |
| 2022.9 - 2023.5 | *Teaching Assistant* | **Florida State University** |
| 2016.9 - 2017.1 | *Exchange Student* | **National Taiwan University** |

## Grants

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| 2025.11 | **NSF NCAR Advanced Study Program** postdoctoral fellowship waitlisted (top 14) |
| 2024.9 - 2026.9 | **$120,000 for 2 years.** Investigating the impact of snow-to-precipitation ratio on surface hydrology in the western United States. |

## Selected Honors and Awards

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| 2026 | Early Career Scientists Best Paper Award, JHU EPS |
| 2025 | AGU Precip Folks, August Highlight |
| 2024 | Glenadore and Howard L. Pim Postdoctoral **Fellowship**, JHU EPS |
| 2023 | **1st place** student oral presentation in the Hydrology section, 103rd AMS |
| 2017 | **National Scholarship** for outstanding undergraduates, $1300 (top 2% in NJU) |
| 2016 | The Liao's Scholarship, $800 (University-level, top 2% in school, NJU) |
| 2015 | The Liao's Scholarship, $800 (University-level, top 2% in school, NJU) |
| 2015 | University-level outstanding students (top 5% in NJU) |
| 2015 | School-level outstanding student leaders (5 out of 83, NJU) |
| 2015 | School-level outstanding students (5 out of 83, NJU) |

## PUBLICATIONS

*Peer-Reviewed Journal Articles* * for corresponding author

6. **Shi, S.***, He, C., Abolafia-Rosenzweig, R., & Zaitchik, B. (2026). Atmospheric Profile Information Improves Rain-Snow Partitioning Accuracy. (Submitted to **Geophysical Research Letters**)
5. **Shi, S.***, Liu, G., and Zaitchik, B. (2025). The climatology and temperature sensitivity of the snow-to-precipitation ratio based on satellite observations (***Water Resource Research***, in review).
4. **Shi, S.*** and Liu, G. (2024). Improvements on phase classification using atmospheric melting and refreezing energy based on soundings. ***Journal of Geophysical Research: Atmospheres***, 129(10), e2023JD040030. https://doi.org/10.1029/2023JD040030.
3. Jeoung, H., **Shi, S.,** & Liu, G.* (2022). A novel approach to validate satellite snowfall retrievals by ground-based point measurements. ***Remote Sensing***, 14(3), 434. https://doi.org/10.3390/rs14030434.
2. **Shi, S.*,** & Liu, G. (2021). The latitudinal dependence in the trend of snow event to precipitation event ratio. ***Scientific Reports***, 11(1), 18112. https://doi.org/10.1038/s41598-021-97451-9.
1. **Shi, S.,** & Misra, V.*. (2020). The role of extreme rain events in Peninsular Florida's seasonal hydroclimate variations. ***Journal of Hydrology***, 589, 125182. https://doi.org/10.1016/j.jhydrol.2020.125182.

*In Preparation*

3. **Shi, S.,** Zhang, Y.*, & Zaitchik, B. An Energetic Classification of Regimes of Precipitation Changes (In preparation).
2. Nie, W., Zaitchik, B., **Shi, S.**, Sleeman, J. A., & Brett, J. AI-Enabled Modeling of Amazon Monsoon Tipping Behavior under Deforestation and Drought (Presented at HydroML workshop).
1. Hu, X., Miao, X.*, Guo, W., **Shi, S.,** Guo, Q., Wang, L., Li, Y. Impact of the Warming-wetting Trend on Snowfall Change over the Tibetan Plateau (To be submitted).

*Open-source Software*

1. **Shi, S.** (2025). ENERGYPHASE, precipitation phase partitioning method incorporated with atmospheric information. https://zenodo.org/records/15033215. GitHub link.

## SERVICE AND LEADERSHIP

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| 2026.4 | JHU EPS Research Day Coordinator |
| 2026 - now | AGU Precipitation Technical Committee, ECSPrecip Seminar Coordinator |
| 2025 - now | Newsletter Editor, Chinese-American Oceanic and Atmospheric Association |
| 2019 - now | Reviewer of *Journal of Hydrometeorology, Climate Dynamics, Journal of Hydrology, Asia-Pacific Journal of Atmospheric Sciences* |
| 2022.8 | Webinar host: Discussion on IPCC AR6 Chap12: Snow and Ice |
| 2015 - 2016 | Group leader of undergraduate student innovative research: Simulating the Double Vortices' Fujiwara Effect in a Rotating Water Tank |

## SKILLS

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| Coding | Python (Scikit-learn, XGBoost), MATLAB, Fortran, C |
| Statistical | Higher moments, Hypothesis testing, Time series analysis, Regressions |
| Modeling | **Noah-MP** land surface model, **HRLDAS**, NASA **LISF**. |
| Lab | Rotating water tank experiments; Snow field measurement school (2027) |